Machine Learning-Based Mixed Refrigerant Optimization to Reduce Compression Power and Fuel Gas Consumption In Badak LNG

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Presented by: PNJ vs everybody



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Presentation Outline



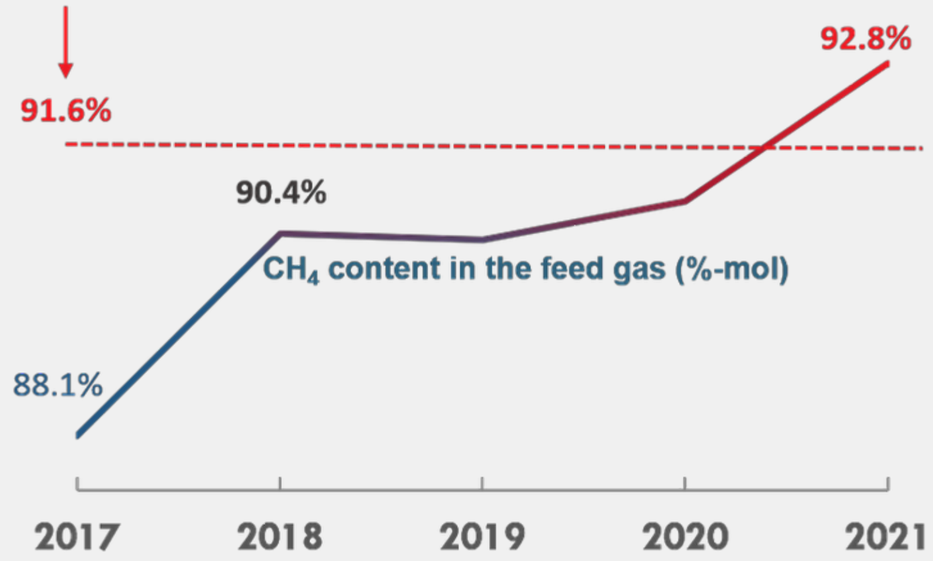
- Introduction
- Methodology
- Results and Discussions
- Conclusion

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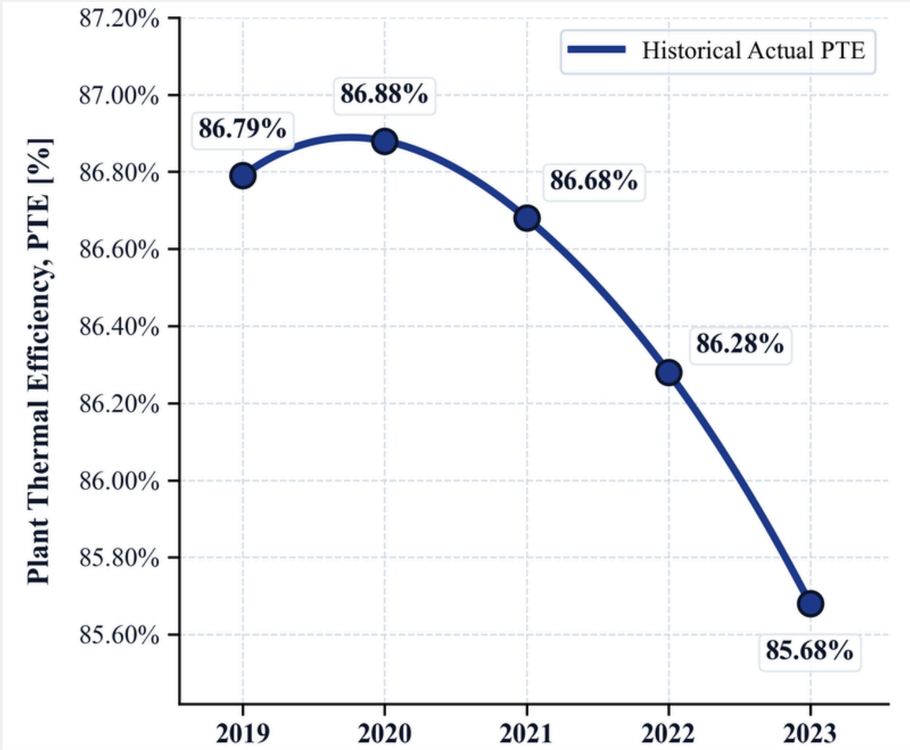
Introduction - Process Needs Analysis

Since 2017, Badak LNG's **rich gas** turned into lean gas

Maximum CH_4 content in the feed gas that can be processed in Bontang LNG Plant (based on Licensor Study in 2019)



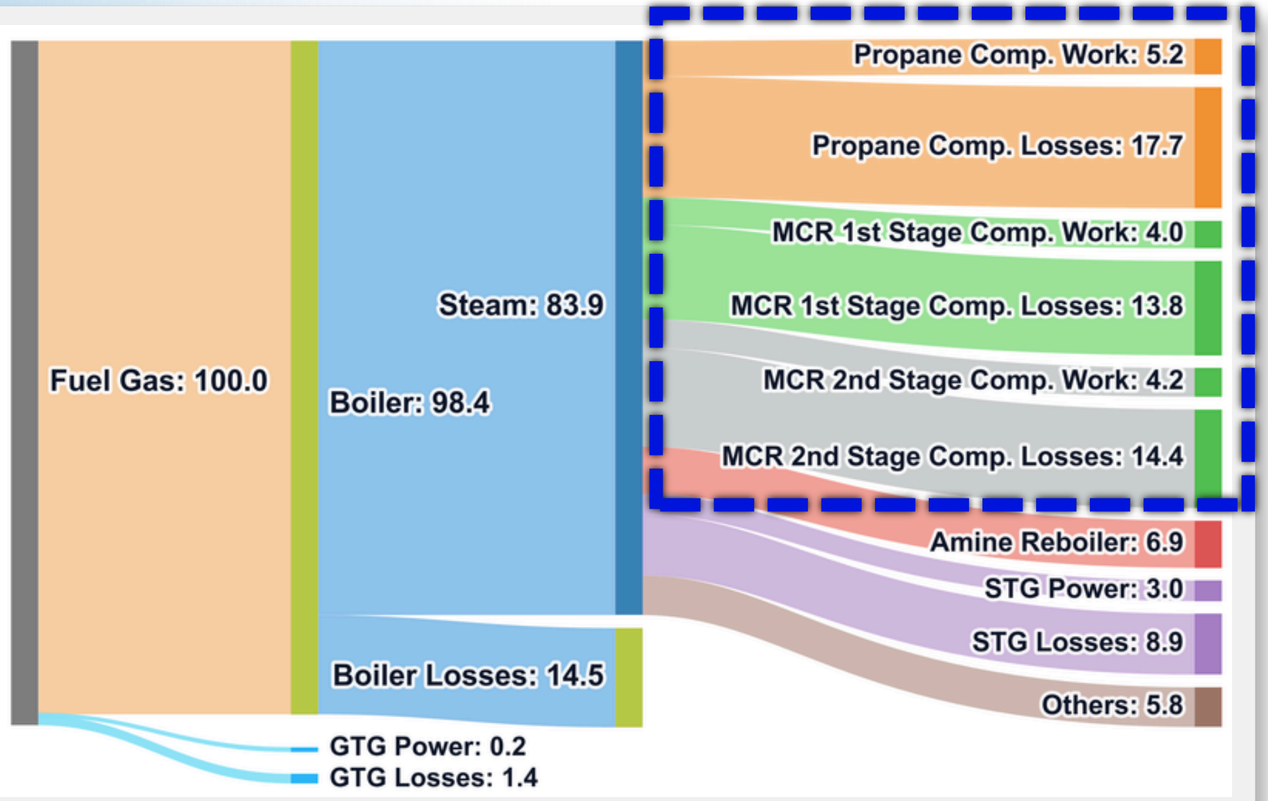
Affecting Plant Thermal Efficiency



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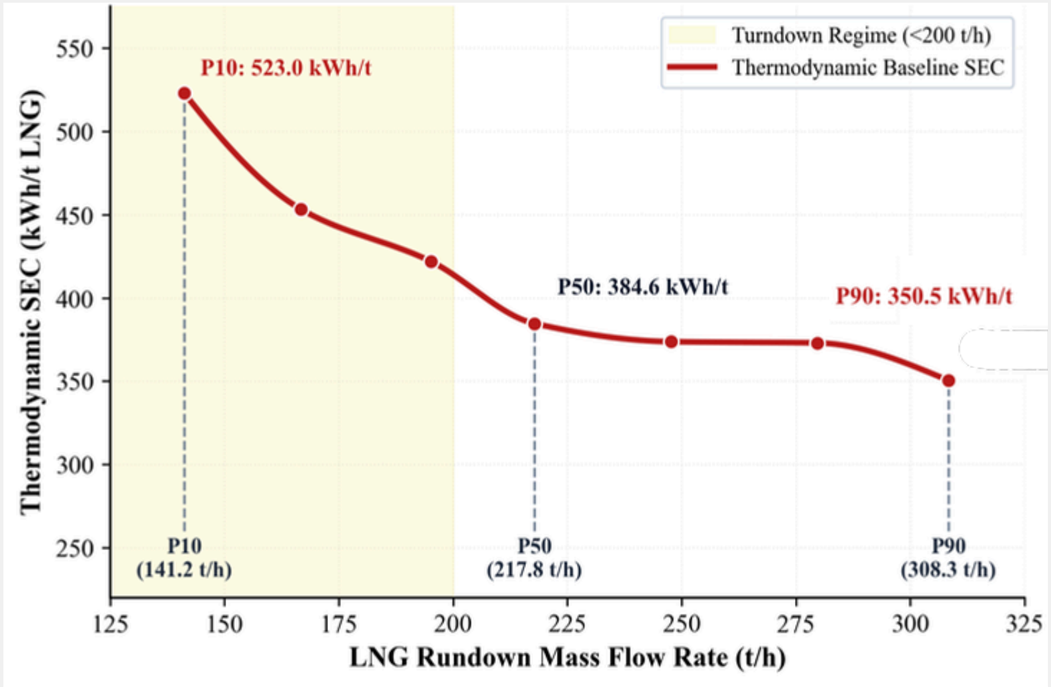
Badak LNG needs a way to improve its PTE

How To Improve Plant Thermal Efficiency?



C3MR contributed 59.3% of Badak LNG's **total energy consumed**

Adjusting **MR composition and flow rate** can reduce **specific energy consumption** of C3MR



How can MR composition and flow rate be optimized across operating loads to reduce SEC while maintaining plant's constraints?

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Methodology - Optimization Framework

Method: five steps from plant data to verified recipe

1 Data Processing

Cleansed **720k+ DCS** records into 7 representative plant loads (P10–P90)

720,734 — **One minute records**, 2025-2026

110,137 — After removing startup, shutdown and idle (LNG rundown > 50 t/h

67,916 — Steady state records for surrogate development

2 Machine Learning Modelling

Trained **HGBR** model to instantly predict compressor power and thermal margins.

3 Mixed Refrigerant Optimization

Adjusted MR flow and composition to **minimize Specific Energy Consumption (SEC).**

4 Thermodynamic Check

Validated MITA, compressor surge, and temperatures rigidly via **Aspen HYSYS.**

5 Impact Assessment

Quantified power reductions into direct fuel gas savings, USD, and avoided CO_2.

A systematic 5 step workflow translating raw plant data into verified economic value.

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Physics Constrained ML & Safety Bounds

Establishing rigorous operational boundaries prior to AI optimization

1

The ML Surrogate

Algorithm: HistGradientBoosting Regressor **(HGBR).**

Architecture: Histogram-based decision tree ensemble for non-linear process dynamics.

Function: Rapid evaluation of total compressor shaft power and thermodynamic margins.

2

Statistical Safety

Space: 4D Isometric Log-Ratio (ILR) coordinates.

Method: Mahalanobis Distance

$$D_M \leq 3.0802$$

95% HISTORICAL TRUST LIMIT

3

Rigorous Validation

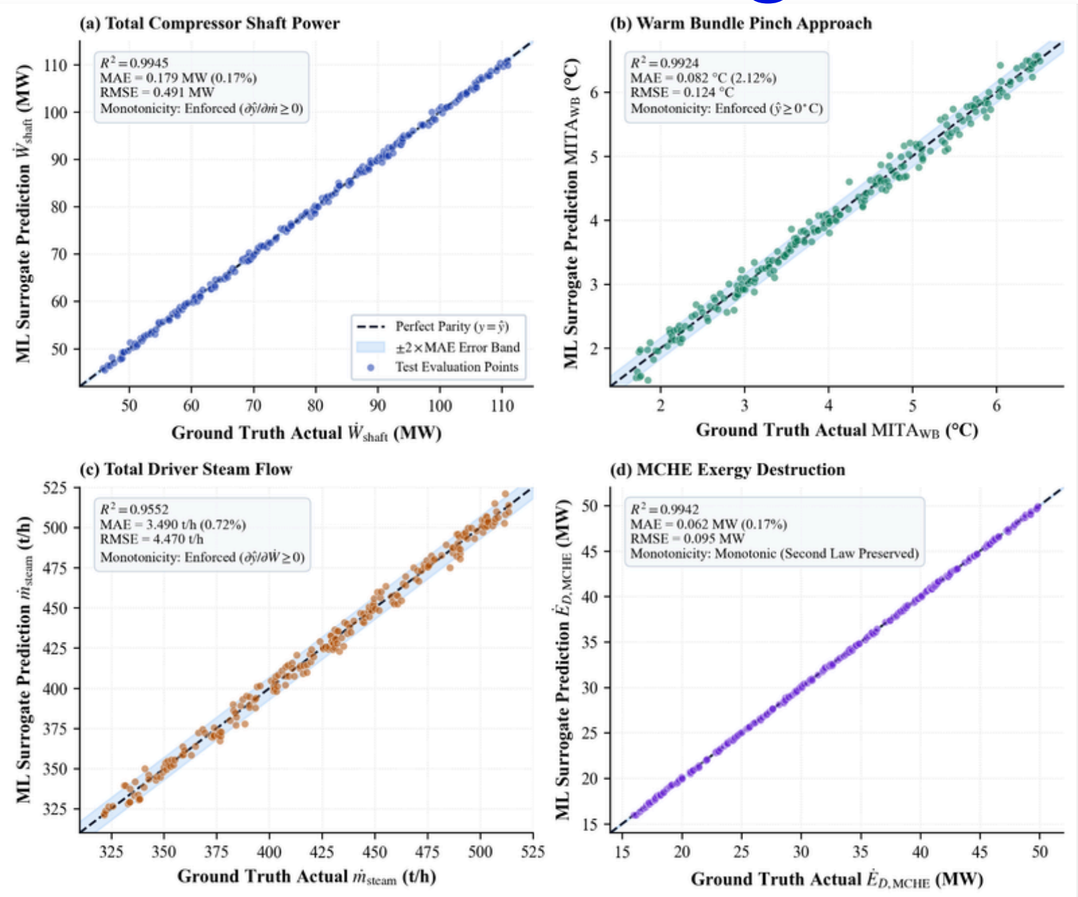
PLANT CONSTRAINT	HARD LIMIT
Anti-Surge Flow	> 480 t/h
Thermal (MITA)	≥ 1.5 °C
MCHE ΔP	< 0.35 kg/cm²
Suction Temp	≥ -45 °C

Validated via **Aspen HYSYS**

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Bridging AI and thermodynamics to establish 100% plant safe operational parameters

Machine Learning Model Validation



Target	Test R^2	Test MAE
Total Compressor Shaft Power (MW)	0.9945	0.179
Warm Bundle Pinch Approach (°C)	0.9924	0.082
Cold Bundle Pinch Approach (°C)	0.9891	0.095
MCHE Exergy Destruction (MW)	0.9942	0.062
Compressor Suction Temp (°C)	0.9664	0.87
Total Driver Steam Flow (t/h)	0.9552	3.49

HGBR surrogates reach **R^2 of 0.9552 to 0.9945** and stay physically consistent, so they are reliable for MR optimization.

Data: 720,734 → 67,916 steady-state records; 2,000 simulation cases

HYSYS check: max deviation 0.14%

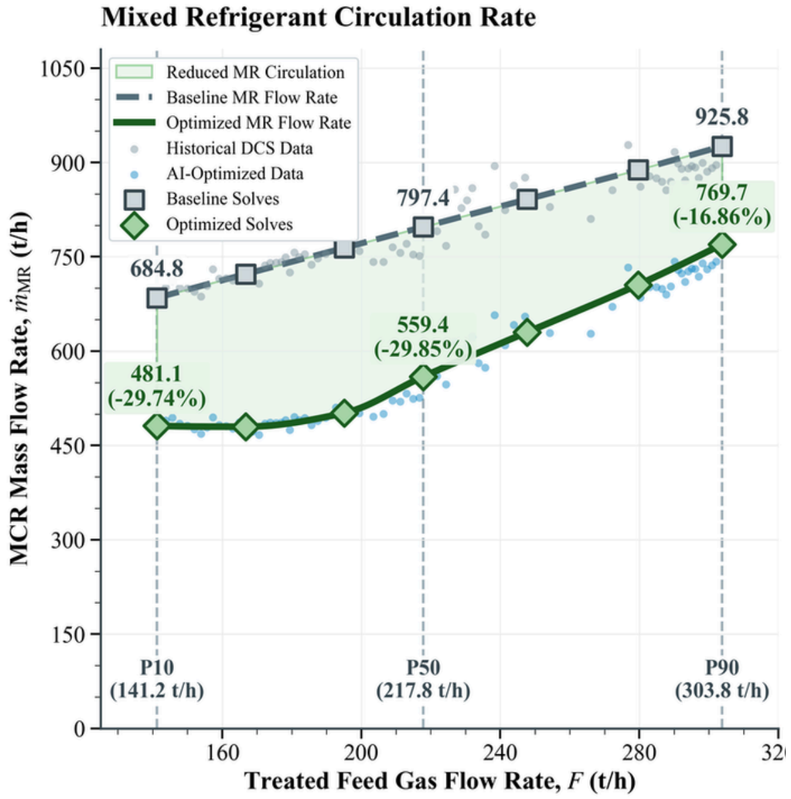
Physics check: 0 negative MITA in 1,000,000 test points

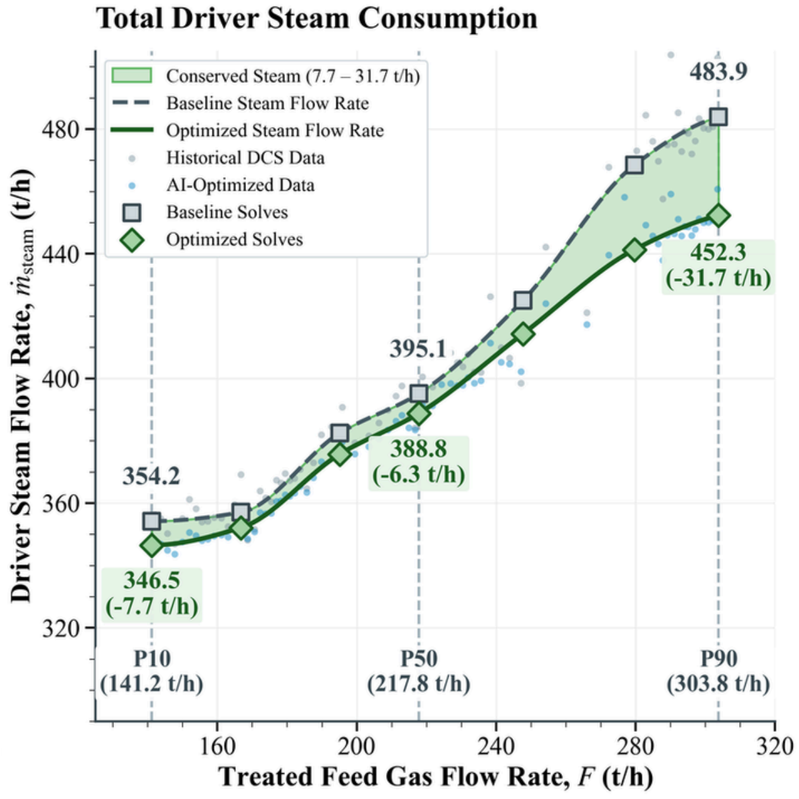
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High accuracy alone is not enough. Surrogates validated against HYSYS (0.14% deviation) and physically consistent (0 negative MITA) are safe to use for optimization.

One MR Recipe, Load-Conditioned Circulation

Nitrogen	Methane	Ethane	Ethylene	Propane	Inside 95% Historical Trust Limit
3.77%	**42.22%**	**43.67%**	**1.53%**	**8.82%**	**D_M = 1.2723 ≤ 3.0802 (limit)**





Baseload performance (P90) after meeting thermodynamic and actual plant's constraints:

925.80 → 769.74 t/h
of MR flow rate reduced

483.95 → 452.30 t/h
of steam flow rate reduced

Optimized MR composition reduces MCHE refrigerant flow and C3MR steam consumption across operating regimes.

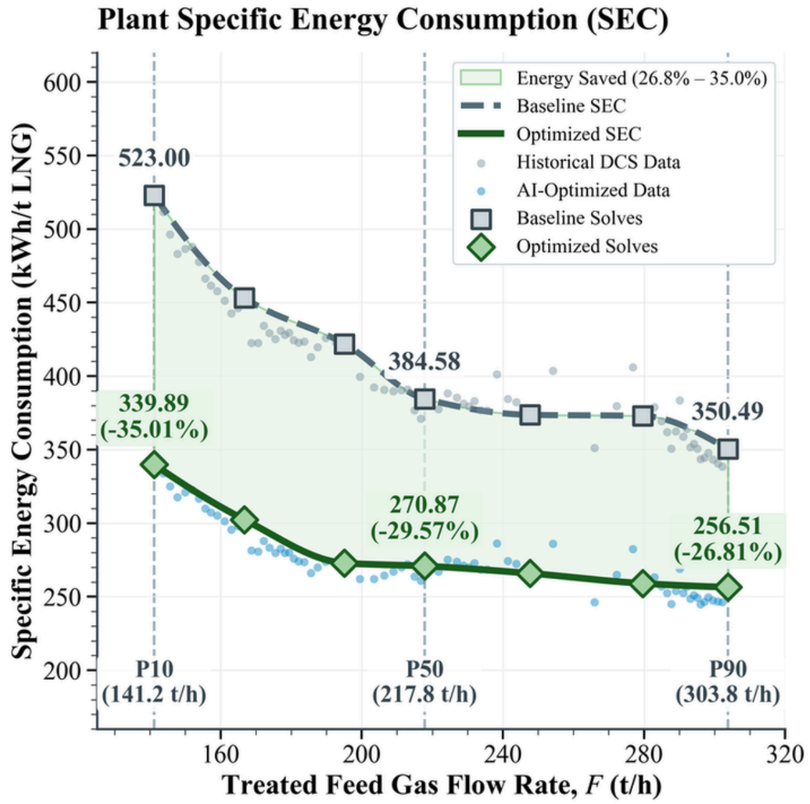
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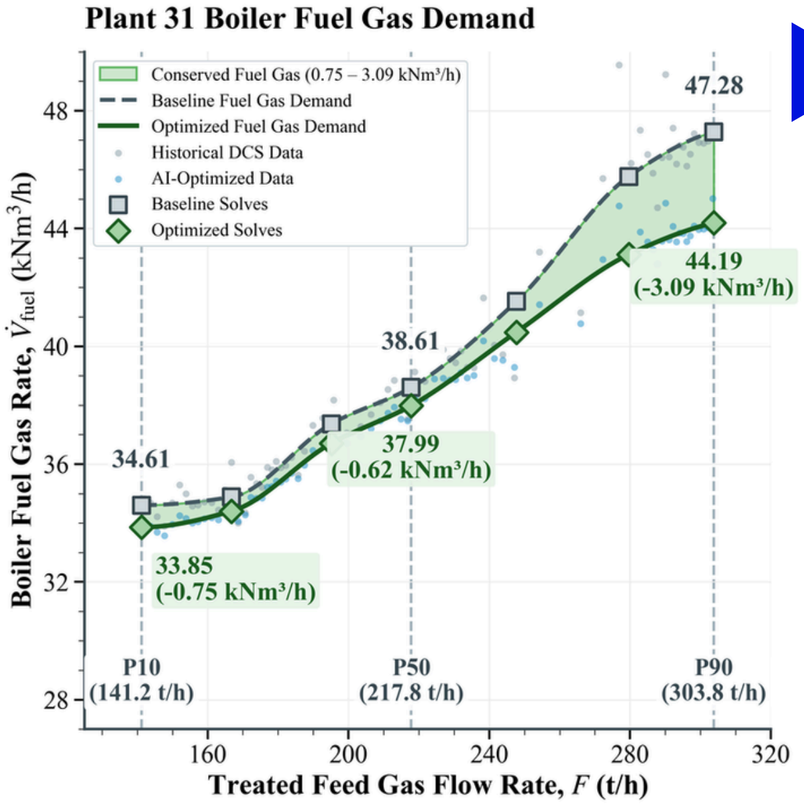
Energy, Economic & Environmental Impact

On baseload performance, **optimization saved:**

-26.81%
of SEC (kWh/t_{LNG})

-3.09%
of Fuel Gas (kNm^3/h)





Weighted by 7,849.9 operating h/y:

239,890 t/y
of turbine steam saved

23.44 M Nm^3/y
of fuel gas saved

49,961.8 t/y
of CO_2e avoided

8.48M USD/year
of operating cost saved

Load-Conditioned MR Optimization Reduces Energy Intensity and Projects Significant Annual Fuel-Gas and CO_2 Savings.

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Conclusion

Delivering measurable value through physics-constrained AI.

Zero CAPEX Implementation

A single optimized MR composition safely accommodates varying LNG production loads simply by adjusting flow rates.

Universal Composition

3.77% N_2 | 42.22% CH_4

43.67% C_2H_6 | 1.53% C_2H_4

8.82% C_3H_8

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of turbine steam saved

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Massive Value Creation

Optimized SEC falls to 256.51 kWh/t LNG at P90 and 339.89 kWh/t LNG at P10.

Massive energy savings achieved without violating plant physics or historical operating boundaries.

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Delivering massive financial and environmental impact (USD 8.48M & 50k tonnes CO_2/y with Zero CAPEX.

Biggest Appreciation & Apology

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